

Anaerobic digestion operating notes

Original sample document for this repository (MIT). Written to exercise retrieval and citations; not a plant design manual.

Temperature

Mesophilic anaerobic digestion typically operates between 35 and 40 deg C. Prolonged drops below 32 deg C reduce methanogenic activity and daily biogas yield. Thermophilic operation (around 50 to 55 deg C) can raise specific gas production but is more sensitive to temperature swings and ammonia inhibition.

pH and alkalinity

The digester pH optimum is 6.8 to 7.5. Values below 6.5 indicate acidification. Reduce the organic loading rate until volatile fatty acids decline and pH restabilizes. Maintain alkalinity so the buffer can absorb short-term acid peaks from a change in feedstock.

Organic loading

Volumetric organic loading rates of 2 to 5 kg VS/m³/day are typical for stable mesophilic continuously stirred tank reactors treating agricultural slurry or energy crops. Overloading leads to volatile fatty acid accumulation, falling methane content, and foaming risk.

Hydraulic retention

For cattle slurry in a mesophilic CSTR, hydraulic retention time is often 20 to 40 days. Shorter retention raises washout risk for slow-growing methanogens. Longer retention improves volatile-solids destruction but increases reactor volume.

Gas quality

Stable plants often produce biogas at 50 to 65% methane by volume, with the balance mainly carbon dioxide plus traces of hydrogen sulfide and water vapor. A sudden drop in methane fraction together with rising VFA is a classic overload signature.